The Subtraction, Explicitly

Griffiths Toy Theory Bubble Integral (Ch. 6, §6.6)

PHYS 7502 — Supplementary Notes

Goal

Take the divergent bubble integral from the toy theory, evaluate it with a cutoff, perform the subtraction at a reference point, and show that the $\ln \Lambda$ cancels — leaving a finite, predictive remainder.

Step 1: The integral

The bubble on the internal C line of $A + A \to B + B$ gives (after using up the delta functions):

$$I(p^2) \;=\; \int \frac{d^4q}{(2\pi)^4}\, \frac{1}{\big[(p-q)^2 - m^2\big]\big[q^2 - m^2\big]}$$

where $p = p_1 - p_3$ is the external momentum flowing through the C line, and for simplicity we set $m_A = m_B = m$ following Griffiths.

This integral **diverges logarithmically** (the integrand goes like $1/q^4$ at large q, while the measure contributes $q^3\, dq$).

Step 2: Feynman parametrize

Feynman parameter identity

$$\frac{1}{AB} = \int_0^1 \frac{dx}{\big[xA + (1-x)B\big]^2}$$

Set $A = (p-q)^2 - m^2$ and $B = q^2 - m^2$. The combined denominator is:

$$\begin{aligned} D &= x\big[(p-q)^2 - m^2\big] + (1-x)\big[q^2 - m^2\big] \\ &= x\big[q^2 - 2p\cdot q + p^2 - m^2\big] + (1-x)\big[q^2 - m^2\big] \\ &= q^2 - 2x\, p\cdot q + x\, p^2 - m^2 \end{aligned}$$

Complete the square by defining the shifted variable $\ell \equiv q - x\, p$:

$$D \;=\; \ell^2 - \underbrace{\big[m^2 - x(1-x)\, p^2\big]}_{\equiv\; \Delta(p^2)}$$

Key definition

$$\Delta(p^2) \;\equiv\; m^2 - x(1-x)\, p^2$$

This is the object that carries all the p^2 dependence. It will survive the subtraction. Everything else will cancel.

After the shift $q \to \ell + xp$, the measure is unchanged ($d^4q = d^4\ell$), so:

$$I(p^2) = \int_0^1 dx \int \frac{d^4\ell}{(2\pi)^4} \frac{1}{\left[\ell^2 - \Delta\right]^2}$$

Step 3: Wick rotate and do the radial integral

Wick-rotate: $\ell^0 \to i\ell_E^0$, so $\ell^2 \to -\ell_E^2$ and $d^4\ell \to i\, d^4\ell_E$. The 4D angular integral gives $2\pi^2$, and $d^4\ell_E = \ell_E^3\, d\ell_E\, d\Omega_4$:

$$I(p^2) = \frac{i \cdot 2\pi^2}{(2\pi)^4} \int_0^1 dx \int_0^\Lambda \frac{\ell_E^3\, d\ell_E}{\left[\ell_E^2 + \Delta\right]^2}$$

The prefactor simplifies: $\dfrac{2\pi^2}{(2\pi)^4} = \dfrac{1}{8\pi^2}$.

The radial integral is standard. Substituting $u = \ell_E^2 + \Delta$:

$$\int_0^\Lambda \frac{\ell^3\, d\ell}{(\ell^2+\Delta)^2} = \frac{1}{2}\Big[\ln(\ell^2+\Delta) + \frac{\Delta}{\ell^2+\Delta}\Big]_0^\Lambda = \frac{1}{2}\left[\ln\frac{\Lambda^2+\Delta}{\Delta} + \frac{\Delta}{\Lambda^2+\Delta} - 1\right]$$

For $\Lambda^2 \gg \Delta$ (which is the regime we care about):

$$\approx \frac{1}{2}\left[\ln\frac{\Lambda^2}{\Delta} - 1\right]$$

Putting it all together:

Regulated result

$$\boxed{I(p^2) = \frac{i}{16\pi^2}\int_0^1 dx\, \left[\ln\frac{\Lambda^2}{\Delta(p^2)} - 1\right]}$$

$$= \frac{i}{16\pi^2}\left[\ln\Lambda^2 - 1 - \int_0^1 dx\, \ln\left[m^2 - x(1-x)\,p^2\right]\right]$$

Structure: The $\ln\Lambda^2$ sits out front with **no p^2 dependence**. The p^2 dependence lives entirely in the second integral, which is finite.

Step 4: The subtraction

Now evaluate the **same** integral at the reference point $p^2 = m_C^2$ (the C mass shell):

$$I(m_C^2) = \frac{i}{16\pi^2}\left[\ln\Lambda^2 - 1 - \int_0^1 dx\, \ln\left[m^2 - x(1-x)\,m_C^2\right]\right]$$

Subtract:

$$I(p^2) - I(m_C^2) = \frac{i}{16\pi^2}\Bigg\{ \Big[\underbrace{\ln\Lambda^2 - 1}_{\text{from } I(p^2)}\Big] - \Big[\underbrace{\ln\Lambda^2 - 1}_{\text{from } I(m_C^2)}\Big]$$
$$- \int_0^1 dx\ \ln\big[m^2 - x(1-x)\,p^2\big]$$
$$+ \int_0^1 dx\ \ln\big[m^2 - x(1-x)\,m_C^2\big]\Bigg\}$$

The first line is $(\ln\Lambda^2 - 1) - (\ln\Lambda^2 - 1) = 0$. **The divergence has cancelled.**

The finite remainder

$$\boxed{I(p^2) - I(m_C^2) \;=\; \frac{i}{16\pi^2}\int_0^1 dx\ \ln\frac{m^2 - x(1-x)\,m_C^2}{m^2 - x(1-x)\,p^2}}$$

- **No Λ anywhere.** The cutoff is gone.
- **Finite.** The argument of the logarithm is a ratio of two positive quantities (in the physical region), so the integral converges.
- **Depends on p^2.** This is the physical, observable correction — how the propagator's behavior at momentum p differs from its behavior at the mass shell.

Step 5: What happened to the divergent piece?

The quantity $I(m_C^2)$ — the piece we subtracted — contains the $\ln\Lambda$ and diverges. But look at how it enters the full amplitude. The tree-level + bubble correction is:

$$\mathcal{M} = \frac{g^2}{p^2 - m_C^2}\left[1 + g^2\, I(p^2)\right] + \cdots$$

Replace $I(p^2) = I(m_C^2) + [I(p^2) - I(m_C^2)]$:

$$\mathcal{M} = \frac{g^2}{p^2 - m_C^2}\Big[1 + g^2\, I(m_C^2)\Big] + \frac{g^4}{(p^2 - m_C^2)^2}\Big[I(p^2) - I(m_C^2)\Big] + \cdots$$

The first bracket is a constant multiplicative correction to the propagator at the mass shell. It redefines the coupling and mass:

$$g^2\big[1 + g^2 I(m_C^2)\big] \;\longrightarrow\; g^2_{\text{phys}}$$

So the final physical amplitude is:

$$\mathcal{M} = \frac{g^2_{\text{phys}}}{p^2 - m^2_{C,\text{phys}}} + \frac{g^4_{\text{phys}}}{(p^2 - m^2_{C,\text{phys}})^2}\underbrace{\big[I(p^2) - I(m_C^2)\big]}_{\text{finite, no }\Lambda} + \cdots$$

Everything is expressed in terms of measurable quantities. Λ is gone.

Step 6: Why the same $\ln \Lambda$ appears at every p^2

The physical reason the subtraction works

The divergence comes from the UV region of the loop: $|q| \gg |p|$, where the loop momentum is so large that the external momentum p is negligible. In that regime, the integrand becomes:

$$\frac{1}{q^2 \cdot q^2} \quad \text{(independent of } p\text{)}$$

Since the divergent part of the integral doesn't know what p is, it contributes the **same** $\ln \Lambda^2$ regardless of the value of p^2. That's why $I(p^2)$ and $I(m_C^2)$ share the same divergence, and why it cancels when you subtract.

This is not a coincidence or a trick. It's a structural feature of loop integrals: UV divergences are **local** (short-distance), so they cannot depend on **long-distance** information like external momenta. The subtraction exploits this locality.

Summary: the three lines that matter

The subtraction in three equations

Before subtraction (divergent):

$$I(p^2) = \frac{i}{16\pi^2}\left[\ln \Lambda^2 - 1 - \int_0^1 dx\ \ln\!\left(m^2 - x(1-x)\,p^2\right)\right]$$

Reference value (equally divergent):

$$I(m_C^2) = \frac{i}{16\pi^2}\left[\ln \Lambda^2 - 1 - \int_0^1 dx\ \ln\!\left(m^2 - x(1-x)\,m_C^2\right)\right]$$

After subtraction (finite):

$$I(p^2) - I(m_C^2) = \frac{i}{16\pi^2}\int_0^1 dx\ \ln\frac{m^2 - x(1-x)\,m_C^2}{m^2 - x(1-x)\,p^2}$$

The $\ln \Lambda^2$ is **identical** in both lines and cancels. What survives is a **finite ratio** inside a convergent integral — and that ratio is the **entire physical content** of the one-loop correction.